

Blockstream

Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-06-24 15:26 UTC

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Risk: Critical

Key Metrics

Pages scanned: 28886

Report type: Premium Executive Audit

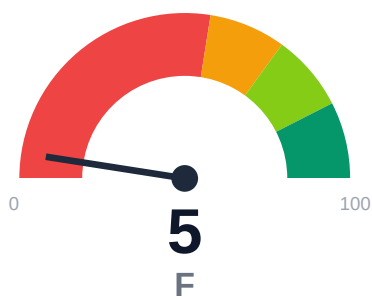
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Crawl coverage is low: 4.02% (30805 URLs examined of 766787 in discovered scope). Run auth
- Broken links: 1222 on docs sites, 190 in repository navigation (repo links are not user-fa
- Link verification inconclusive for 682657 URLs (auth/rate-limit/server block); these are e
- SEO/GEO issue rate is high: 27.8%

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit across 9 Blockstream properties crawled 28,886 pages (4.02% of 766,787 discovered URLs). 1,222 broken links exist in user-facing docs (190 additional broken repo navigation links). API reference coverage is 14.57% across 984 detected API pages. Retrieval readiness scores 64.89/100 (moderate band) with only 53.3% answerability and 62.53% stale-risk flagged pages. Only 37.74% of pages display last-updated metadata, and 27.8% have SEO/geo issues.

External posture: SEO/GEO issue rate **27.8%**, public API coverage **14.6%**, broken links **1222**.

5 Audit Score	F Grade	Critical Risk Band
28886 Pages Scanned	1222 Broken Links	27.8% SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://docs.liquid.net/	0	0	N/A	0.0	0.0
https://docs.corelightning.org/	0	0	N/A	0.0	0.0
https://gdk.readthedocs.io/	36	21	N/A	0.0	50.0
https://blockstream.com/satellite-api-document	27	0	23.5	0.0	26.9
https://wally.readthedocs.io/	12	62	N/A	0.0	50.0
https://blockstream.github.io/docs	1	0	N/A	0.0	25.0
https://elementsproject.org/	390	0	0.0	0.0	1.4
https://help.blockstream.com/	335	13	42.6	93.3	50.5
https://github.com/Blockstream/esplora/blob/m	28085	1316	5.8	100.0	27.9

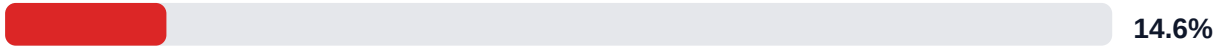
Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-87
Industry average (developer docs)	85	-80
Minimum acceptable	70	-65
Your score	5	-

Gap to industry average: **80 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public)



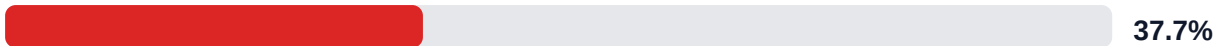
SEO/GEO Issue Rate (lower is better)



Example Reliability (estimate)

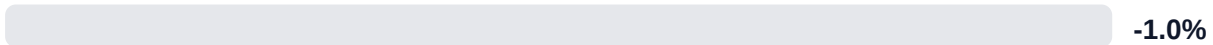


Freshness Visibility (last-updated metadata)

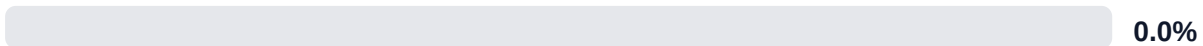


Internal Metrics (from scorecard)

API Coverage (Internal)



Example Reliability (Internal)



Docs-Contract Drift



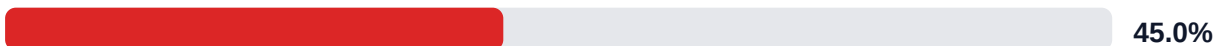
Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.

LLM Retrieval Readiness (deterministic proxy)

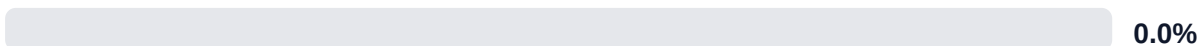


Audit Confidence (crawl scope + verification quality)



Readiness band: **Moderate**. Confidence: **Low**. Open remediation opportunities: **4**.

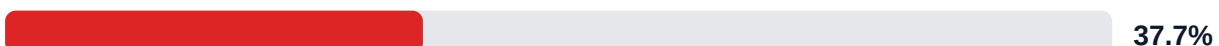
SEO/GEO Quality Signal (inverse issue rate)

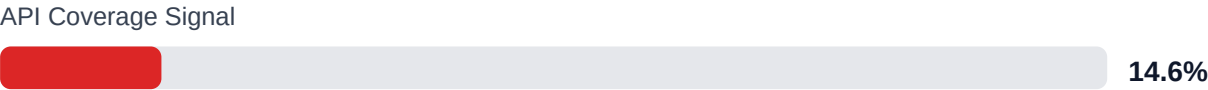


Example Reliability Signal

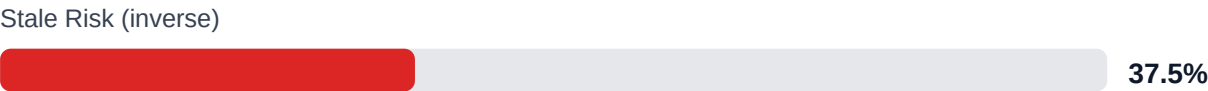


Freshness Visibility Signal





LLM retrieval readiness breakdown (deterministic)



Evidence coverage: 51.6% (17595/34065 key claims backed by deterministic evidence signals).

Highest-value remediation targets

Problem	Severity	Pipeline solution path
Internal documentation link integrity issues	HIGH	Weekly link governance via automated audits and consolidated remediation queue
Search and AI readability quality gaps	HIGH	Deterministic SEO/GEO linting and fix cycles before publish
Weak freshness and lifecycle visibility	MEDIUM	Lifecycle management, stale detection, and weekly governance cadence
API reference coverage and drift risk	HIGH	API-first contract flow, validators, regression gates, and drift checks

Financial Exposure Model

Low Estimate \$392,317 annual exposure Remediation: \$1,805 Monthly loss: \$3,562 Opportunity: \$28,980/mo	Base Estimate \$486,187 annual exposure Remediation: \$4,702 Monthly loss: \$11,144 Opportunity: \$28,980/mo	High Estimate \$643,554 annual exposure Remediation: \$7,600 Monthly loss: \$24,016 Opportunity: \$28,980/mo
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Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-API-COVERAGE	Undocumented API operations	HIGH	\$2,508	\$950
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH	\$1,938	\$808
F-FRESHNESS	Documentation freshness debt	HIGH	\$969	\$475
F-DRIFT	Code/docs contract drift	HIGH	\$1,642	\$570
F-LAYERS	Missing required doc layers	HIGH	\$1,539	\$712
F-RETRIEVAL	RAG retrieval quality risk	HIGH	\$1,949	\$855
F-TERMINOLOGY	Terminology inconsistency	LOW	\$598	\$332

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-API-COVERAGE	Undocumented API operations	HIGH
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH
F-FRESHNESS	Documentation freshness debt	HIGH
F-DRIFT	Code/docs contract drift	HIGH
F-LAYERS	Missing required doc layers	HIGH

Recommended Actions

1. Fix 1,222 broken internal documentation links across user-facing sites [impact: critical, effort: medium]
2. Increase API reference coverage from 14.57% to at least 60% by documenting missing endpoints across 984 detected API pages [impact: critical, effort: high]

3. Add last-updated timestamps to the 62.47% of pages missing freshness metadata (18,000+ pages) [impact: high, effort: low]
4. Improve answerability from 53.3% to 75%+ by restructuring pages with missing context, unclear headings, or incomplete explanations [impact: high, effort: high]
5. Resolve 27.8% SEO/geo issues (8,000+ pages) to improve organic discoverability and regional access [impact: high, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Example code reliability estimated at 98.31%, indicating high quality of working samples
- + Actionability coverage at 98.67% shows docs provide clear next steps
- + Citationability at 99.95% enables accurate source attribution for AI and support
- + Chunkability at 86.11% supports effective search and retrieval systems
- + Multi-project topology correctly identified; no inappropriate fragmentation penalty

Risks

- 682,657 links unverified due to auth/rate-limit blocks; true broken link count may be significantly higher than reported 1,222
- Crawl reached only 4.02% of discovered scope (28,886 of 766,787 pages); large documentation surface remains unaudited
- Overall audit confidence at 49.1% and link confidence at 25% limit reliability of findings
- API coverage at 14.57% leaves 85% of API surface undocumented or undetected, blocking developer onboarding
- Answerability at 53.3% means nearly half of pages cannot effectively answer user questions in RAG/AI systems
- 62.53% stale-risk flagged pages without freshness signals erode trust and may contain outdated technical guidance
- 27.8% SEO/geo issue rate reduces discoverability in search engines and regional access

Limitations

- * Crawl reached only 4.02% of discovered scope; findings may not represent quality across 738,000+ uncrawled pages, especially behind authentication or dynamic navigation.
- * 682,657 links could not be verified due to auth, rate limits, or server blocks; actual broken link count may be substantially higher than the 1,412 confirmed.
- * Overall confidence at 49.1% and link confidence at 25% indicate significant detection gaps; manual validation required for business-critical decisions.
- * External audit cannot assess accuracy of technical content, code example functionality in user environments, or alignment with current product behavior.
- * Multi-site topology and potential auth-gated content may hide internal inconsistencies, duplicate content, or conflicting guidance across properties.

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	95.0
support_hourly_usd	65.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	6.0
avg_release_delay_hours	18.0
monthly_support_tickets	420.0
docs_related_ticket_share	0.28
avg_ticket_handling_minutes	22.0
monthly_doc_influenced_evals	45.0
eval_to_customer_rate	0.18
avg_customer_monthly_value_usd	3200.0
docs_friction_revenue_sensitivity	0.35

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	MEDIUM
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH

F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	LOW
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All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.